

FACULTY OF ENGINEERING

B.E. (EEE/EIE/ECE/M/P/AE/AI&DS/AI&ML/IoT/IT) II -Semester (AICTE) (Main& Backlog) (New)
Examination, September/ October 2023

Subject: Programming for Problem Solving

Time: 3 Hours

Max. Marks: 70

Note: (I) First question is compulsory and answer any four questions from the remaining six questions. Each questions carries 14 Marks.

(II) Answer to each question must be written at one place only and in the same order as they occur in the question paper.

(III) Missing data, if any, may be suitably assumed.

1. ☒ a) Differentiate Compiler and Interpreter.
☒ b) What is the use of break statement?
c) Write an Algorithm to perform Linear Search.
☒ d) What is recursion?
e) What is pointer to pointer?
☒ f) Write a for loop to print 30, 27, 24,3.
g) Illustrate the difference between = and ==.
2. ☒ a) Draw and Explain in detail various components of the computer,
☒ b) Draw the flowchart to find roots of the Quadratic Equation,
3. ☒ a) Write in detail if, if-else, switch with syntax and example.
☒ b) Explain the following string manipulating function strcpy(), strlen and strcat().
4. ☒ a) Sort the numbers using selection sort: 12, 23, 56, 48, 21, 33, 10
b) How is call by value different from call by address.
5. a) What is Structure. Demonstrate with Student example.
☒ b) Write a recursive function to print Fibonacci series.
6. ☒ a) What is Linked List. Draw and represent it.
b) Write a program to read from one file and write to another file.
7. ☒ a) Explain in detail Arithmetic, Relational and Logical Operator.
b) Write a program to read and print a matrix.

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